

Financial Econometrics

Quasi experiments and Difference-in-Difference approach

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Over the past decade, hundreds of exchange listed securities have traded more volume in off-exchange markets than on exchange markets. There has been speculation that the exchange access fee structure might have driven trading away from public lit markets. To test this question, NASDAQ implemented an access fee pilot which reduced the maker rebate and taker fee. In short, maker/taker fees structures charges takers of liquidity a fee and provides a rebate to traders who provide liquidity. In particular, NASDAQ informed the Security Exchange Commission that, *"in response to claims that public markets are too expensive, NASDAQ will modify NASDAQ Rule 7018(a) to lower execution fees for a selected group of securities where access fees may be discouraging the use of public markets."* On February 2, 2015, NASDAQ implemented the maker-taker fee pilot for 14 traded stocks on NASDAQ for which the taker fee was reduced from the market wide level of 0.30 cents to 0.05 cents. Similarly, the maker fee was reduced from 0.29 cents to 0.04 cents. Note that the net effect is tiny. The tickers of the treated stocks which are included in the provided sample are: AAL, BAC, FEYE, GE, GPRO, GRPN, KMI, MU, S, and TWTR.

The NASDAQ access fee reduction pilot offers an informative natural experiment which we can exploit empirically. In this multi-step exercise, you investigate if a reduction in exchange fee structure impacted trading activity. Our measure of trading activity will be the abnormal daily volume. It is defined as daily trading volume relative to daily trading volume average during the period July to November 2014.

Two separate data files are provided allowing you to explore the aforementioned question. In specific, you may find following variables

- *FEE_vol.csv*: daily abnormal volume data from December 2014 to July 2015
 - date: date in the format YYYYMMDD
 - TICKER: stock identifier
 - AVOL: abnormal trading volume, multiply by 100 to display as percentages
- *FEE_char.csv*: monthly averages of the firm characteristics as of October 2014
 - TICKER: stock identifier
 - MCAP: natural logarithm of market cap
 - RNG: proxy for volatility based on the difference between high and low price during the day
 - PRC: inverse of price

Your analysis proceeds in following steps

Part 1: Estimate a naive treatment effect

- Load the data and familiarize yourself with it. Please winsorize all relevant variables used at the 99% level.
- You are interested in whether firms included in the experiment exhibit a change in abnormal trading volume around the policy change. You start with a visual inspection and display average daily AVOL for all treated firms.
- You want to implement a formal test. Estimate a naive treatment effect using following regression

$$AVOL = \alpha + \beta \times AFTER + u$$

- *AFTER* is a dummy variable turning 1 after the fee structure was changed on February 2, 2015. You decide to use following estimation windows: December 2014 to March 2015.
- Use only the firms treated in the experiment and estimate following two specifications which you report as a single stargazer table: (1) no controls, and (2) no controls with firm fixed effects. Use clustered standard errors. Note, we will extend this table throughout the exercise.
- Interpret
 - Did the policy affect trading activity?
 - Is the effect economically significant?
 - Does the inclusion of firm fixed effects matter?

Part 2: Use a Difference in Difference approach (DiD)

- You worry that other unobserved changes around the policy change might have impact on your results. If that was the case, a DiD approach will help. Including other firms in the sample, that were not exposed to the policy change, and assuming that the parallel trend assumption is valid will account for the aforementioned unobserved changes around the event date.
- You are interested in whether firms included in the experiment exhibit a change in abnormal trading volume around the policy change different to all other firms. You start with a visual inspection and compute average daily AVOL for all treated and non-treated firms. Display the difference over time.
- Estimate following regression

$$AVOL = \alpha + \beta_1 \times TREAT + \beta_2 \times AFTER + \beta_3 \times TREAT \times AFTER + u$$

- *TREAT* is a dummy variable equal to 1 for all firms affected by the fee structure.
- In contrast to part 1, use all firms and estimate two specifications (one without and one with firm and time fixed effects) which you add to the stargazer table. Use clustered standard errors.
- Interpret
 - Did the policy affect trading activity?
 - Is the effect economically significant?
 - Does the inclusion of fixed effects matter?

Part 3: Comparing treatment and control group

- You start worrying that including all firms as control firms might be a bit excessive. In particular you worry about random assignment. You are uncertain if NASDAQ choose the treatment firms randomly, or whether they hand picked them. Given the former, using all firms as treatment firms should not include a bias. However, if the latter is true, you introduced a sample selection bias in part 2 and your estimates might be biased.
- Your strategy is simple, you use firm characteristics from October 2014. You run four logit regressions explaining whether or not a firm received treatment (was part of the experiment): (1)-(3) you regress a treatment dummy on individual firm characteristics, and (4) you use all characteristics jointly in one regression. Please make sure that you winsorised the firm characteristics at the 99% level.
- Interpret
 - Are your worries dispersed or not?

Part 4: Constructing a control group based on propensity score matching

- In this task, we select control firms that are same as likely to receive treatment, based on some observable characteristics (Model 4, previous part). We use a logit model explaining treatment with *MCAP*, *PRC*, and *RNG*.

$$Prob(y = 1|MCAP, PRC, RNG) = \Lambda(\alpha + \beta_1 MCAP + \beta_2 PRC + \beta_3 RNG)$$

- For each of the treated firms select three control firms with a similar propensity of being treated given the model in task 1. You might want to use the *MatchIt* package and the *matchit* function. The relevant syntax is *matchit(y ~ x, data = , method = "nearest", distance = "logit", ratio = 3)*. Inspect the output of *matchit* and you will see that *..\$match.matrix* provides the row names of the selected control firms.
- You start with a visual inspection and compute average daily AVOL for all treated and chosen control firms. Display the difference over time. The difference is noisy, so you might want to smooth it.
- After deciding on the appropriate control sample, estimate following regression

$$AVOL = \alpha + \beta_1 \times TREAT + \beta_2 \times AFTER + \beta_3 \times TREAT \times AFTER + u$$

- *TREAT* is a dummy variable equal to 1 for all firms affected by the fee structure.
- In contrast to part 2, use only treated and control firms, and estimate two specifications (one without and one with firm and time fixed effects) which you add to the stargazer table. Use clustered standard errors.
- Interpret
 - Did the policy affect trading activity?
 - Is the effect economically significant?
 - Does the construction of the control sample matter?

```

#-----#
#           SOLUTION           #
#-----#

# packages needed
require(stargazer)
require(DescTools)
require(plm)
require(lmtest)
require(sandwich)
require(ggplot2)
require(dplyr)
require(MatchIt)

#-----#
# Part 1: Estimate a naive treatment effect

# load the data
vol<- read.csv("C:/Users/s13163/Dropbox/FIE401/data/data_labs/FEE_vol.csv")

# additional variables
vol$date.v2 <- as.Date(as.character(vol$date), format = "%Y%m%d")
vol$after <- vol$date > 20150202
vol$treat <- vol$TICKER %in% c("AAL", "BAC", "FEYE", "GE", "GPRO",
                             "GRPN", "KMI", "MU", "S", "TWTR")
vol$window <- vol$date %in% 20141200:20150399

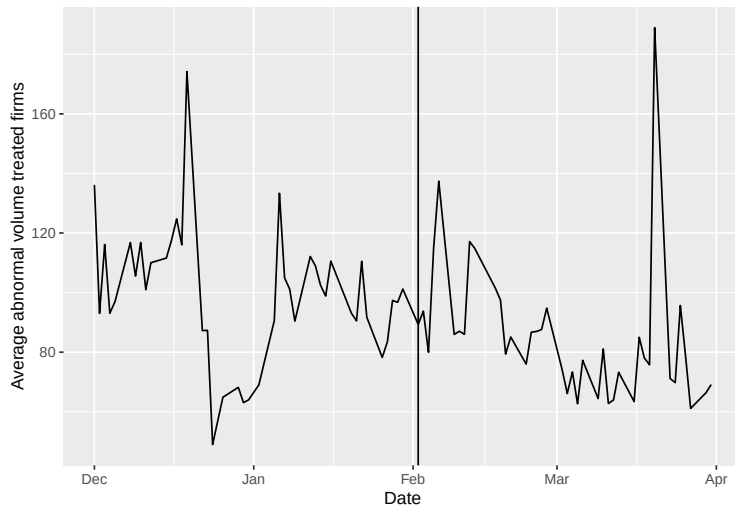
# winsorize
vol$AVOL <- Winsorize(vol$AVOL, val=quantile(vol$AVOL,probs = c(0.005, 0.995),na.rm=TRUE))*100

# declare panel data frame
vol_treat_only<-subset(vol,treat & window)
vol_treat_only<- pdata.frame(vol_treat_only, index = c("TICKER", "date"))
pdim(vol_treat_only)

## Balanced Panel: n = 10, T = 83, N = 830

# Visual inspection
vol_treat_only %>%
  group_by(date.v2) %>%
  summarise(av.AVOL = mean(AVOL)) %>%
  # plot
  ggplot(data = .,
         aes(x = date.v2,
             y = av.AVOL)) +
  geom_line() +
  geom_vline(xintercept = as.Date("2015-02-02")) +
  xlab("Date") +
  ylab("Average abnormal volume treated firms")

```



```
# Estimating the treatment effect model
m1 <- plm(AVOL ~ after, data = vol_treat_only ,
          model= "pooling")
m1.se <- coeftest(m1, vcov =
                  vcovHC(m1, cluster = "group",type="sss"))[,2]

m2 <- plm(AVOL ~ after, data = vol_treat_only,
          model= "within")
m2.se <- coeftest(m2,
                  vcov = vcovHC(m2, cluster = "group",type="sss"))[,2]

# standard deviation of volume for treated firms
sd(vol_treat_only$AVOL)
```

```
## [1] 62.06689
```

```
# output
stargazer(list(m1, m2),
          se = list(m1.se, m2.se),
          type="text", keep.stat=c("n"), report=('vc*t'))
```

```
##
## =====
##               Dependent variable:
##               -----
##                   AVOL
##               (1)           (2)
## -----
## after           -13.875*      -13.875*
##                  t = -1.719    t = -1.720
##
## Constant        99.254***
##                  t = 9.546
##
## -----
## Observations    830           830
## =====
## Note:           *p<0.1; **p<0.05; ***p<0.01
```

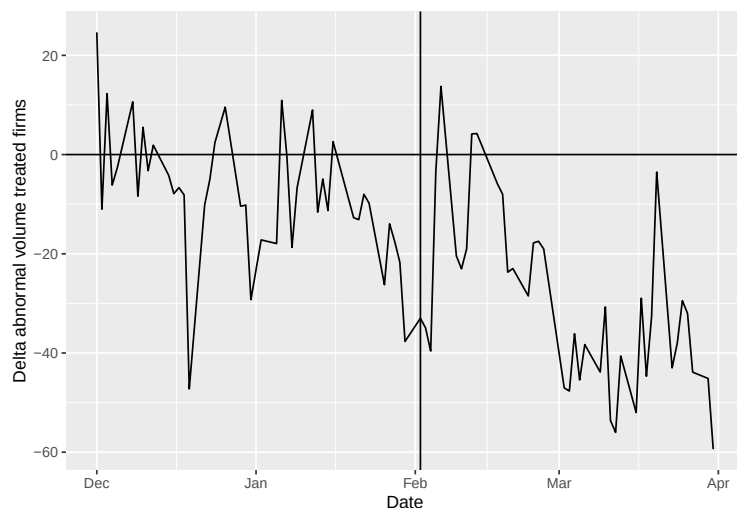
```
#-----#
# Part 2: Use a Difference in Difference approach (DiD)
```

```
# declare panel data frame
vol<-subset(vol, window)
vol<- pdata.frame(vol, index = c("TICKER", "date"))
pdim(vol)
```

```
## Unbalanced Panel: n = 3054, T = 2-83, N = 250359
```

```
# Visual inspection: difference in AVOL
```

```
vol %>%
  group_by(date.v2, treat) %>%
  summarise(av.AVOL = mean(AVOL)) %>%
  ungroup() %>%
  group_by(date.v2) %>%
  summarise(d.AVOL = diff(av.AVOL)) %>%
  # plot
  ggplot(data = .,
    aes(x = date.v2,
        y = d.AVOL)) +
  geom_line() +
  geom_vline(xintercept = as.Date("2015-02-02")) +
  geom_hline(yintercept = 0) +
  xlab("Date") +
  ylab("Delta abnormal volume treated firms")
```



```
# Estimate a Dif-in-Dif using all stocks available
```

```
m3 <- plm(AVOL ~ treat*after, data = vol ,
  model= "pooling")
m3.se <- coeftest(m3, vcov =
  vcovHC(m3, cluster = "group",type="sss"))[,2]

m4 <- plm(AVOL ~ treat*after, data = vol,
  model= "within", effect = "twoway")
m4.se <- coeftest(m4, vcov =
  vcovHC(m4, cluster = "group",type="sss"))[,2]
```

```
# output
stargazer(list(m1, m2, m3, m4),
           se = list(m1.se, m2.se, m3.se, m4.se),
           type="text", keep.stat=c("n"), report=('vc*t'))

##
## =====
##                               Dependent variable:
##                               -----
##                               AVOL
##                               (1)      (2)      (3)      (4)
## -----
## treat                               -8.454
##                               t = -0.854
##
## after      -13.875*   -13.875*   7.159***
##            t = -1.719 t = -1.720 t = 6.559
##
## treatTRUE:after      -21.034*** -21.364***
##                       t = -2.721 t = -2.764
##
## Constant      99.254***      107.708***
##                t = 9.546      t = 115.757
## -----
## Observations      830      830      250,359      250,359
## =====
## Note:                               *p<0.1; **p<0.05; ***p<0.01
```

```
#-----#
# Part 3: Comparing treatment and control group

# load the data
char <- read.csv("C:/Users/s13163/Dropbox/FIE401/data/data_labs/FEE_char.csv")

# construct a treatment dummy
char$treat <- char$TICKER %in%
  c("AAL", "BAC", "FEYE", "GE", "GPRO", "GRPN", "KMI", "MU", "S", "TWTR")

# winsorize
char$MCAP <- Winsorize(char$MCAP, val=quantile(char$MCAP, probs = c(0.005, 0.995), na.rm=TRUE))
char$PRC <- Winsorize(char$PRC, val=quantile(char$PRC, probs = c(0.005, 0.995), na.rm=TRUE))
char$RNG <- Winsorize(char$RNG, val=quantile(char$RNG, probs = c(0.005, 0.995), na.rm=TRUE))

# logit
fit.logit.1 <- glm(treat ~ MCAP, data = char, family = "binomial")
fit.logit.2 <- glm(treat ~ PRC, data = char, family = "binomial")
fit.logit.3 <- glm(treat ~ RNG, data = char, family = "binomial")
fit.logit.4 <- glm(treat ~ MCAP + PRC + RNG, data = char, family = "binomial")

# output
stargazer(list(fit.logit.1, fit.logit.2, fit.logit.3, fit.logit.4),
           type="text", keep.stat=c("n"), report=('vc*t'))

##
## =====
```

```
##                               Dependent variable:
##                               -----
##                               treat
##                               (1)      (2)      (3)      (4)
## -----
## MCAP          0.874***                1.926***
##                t = 4.546                t = 5.276
##
## PRC                    1.273                34.989***
##                t = 0.181                t = 3.755
##
## RNG                    24.074    129.944***
##                t = 1.085    t = 3.785
##
## Constant      -7.124***   -5.790***   -6.276***   -14.181***
##                t = -10.699 t = -11.234 t = -9.764 t = -6.623
## -----
## Observations    3,054      3,054      3,054      3,054
## =====
## Note:                               *p<0.1; **p<0.05; ***p<0.01
```

```
#-----#
# Part 4: Constructing a control group based on propensity score matching

# assign tickers as row names
rownames(char) <- char$TICKER

# using the MatchIt package
matching <- matchit(treat ~ MCAP + PRC + RNG, data = char,
                    method = "nearest", distance = "logit", ratio = 3)

# inspect the output
summary(matching)
```

```
##
## Call:
## matchit(formula = treat ~ MCAP + PRC + RNG, data = char, method = "nearest",
##         distance = "logit", ratio = 3)
##
## Summary of Balance for All Data:
##           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
## distance      0.1375      0.0028      0.7285    165.8526    0.4605
## MCAP          3.0699      0.0624      1.9505     0.7127    0.3969
## PRC           0.0579      0.0554      0.0443     1.6825    0.0794
## RNG           0.0252      0.0214      0.3666     0.8865    0.1370
##           eCDF Max
## distance      0.8156
## MCAP          0.6779
## PRC           0.2122
## RNG           0.3072
##
## Summary of Balance for Matched Data:
##           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
## distance      0.1375      0.0828      0.2958     4.8981    0.0008
```

```
## MCAP          3.0699      2.6139      0.2957      0.3287      0.1556
## PRC           0.0579      0.0605     -0.0472      0.8605      0.1437
## RNG           0.0252      0.0291     -0.3732      0.2072      0.1596
##              eCDF Max Std. Pair Dist.
## distance      0.2000      0.2980
## MCAP          0.3333      1.3571
## PRC           0.3333      1.0884
## RNG           0.3000      1.8081
##
## Sample Sizes:
##              Control Treated
## All           3044      10
## Matched       30       10
## Unmatched     3014      0
## Discarded      0       0
```

```
# find the assigned matched sample
matching$match.matrix
```

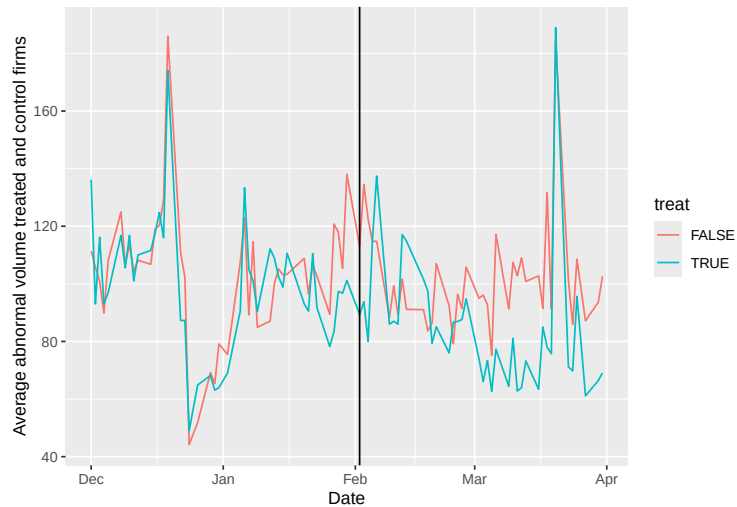
```
##      [,1] [,2] [,3]
## AAL  "PG"  "AMGN" "ESI"
## BAC  "GILD" "PFE"  "FMSA"
## FEYE "STT"  "SGMS" "AMCC"
## GE   "MNKD" "FB"   "ORCL"
## GPRO "SBAC" "MERC" "SZMK"
## GRPN "GRBK" "MRK"  "JPM"
## KMI  "USB"  "VTAE" "UTX"
## MU   "CTCM" "EPE"  "PM"
## S    "CLF"  "INTC" "MSFT"
## TWTR "PEP"  "BIIB" "APC"
```

```
# make control dummy for volume and characteristics data
vol$control.PS <- vol$TICKER %in% as.vector(matching$match.matrix)
```

```
vol_match<-subset(vol, treat | control.PS)
```

```
# Visual inspection: average AVOL over time both groups
```

```
vol_match %>%
  group_by(date.v2, treat) %>%
  summarise(av.AVOL = mean(AVOL)) %>%
  # plot
  ggplot(data = .,
    aes(x = date.v2,
      y = av.AVOL,
      color = treat)) +
  geom_line() +
  geom_vline(xintercept = as.Date("2015-02-02")) +
  xlab("Date") +
  ylab("Average abnormal volume treated and control firms")
```



```
# Estimate a Dif-in-Dif using matched sample
m5 <- plm(AVOL ~ treat*after, data = vol_match,
          model= "pooling")
m5.se <- coeftest(m5, vcov =
                  vcovHC(m5, cluster = "group", type="sss"))[,2]

m6 <- plm(AVOL ~ treat*after, data = vol_match,
          model= "within", effect = "twoway")
m6.se <- coeftest(m6, vcov =
                  vcovHC(m6, cluster = "group", type="sss"))[,2]

# output
stargazer(list(m1, m2, m3, m4, m5, m6),
          se = list(m1.se, m2.se, m3.se, m4.se, m5.se, m6.se),
          type="text", keep.stat=c("n"), report=('vc*t'))
```

```
##
## =====
##                               Dependent variable:
##                               -----
##                               AVOL
##                               (1)      (2)      (3)      (4)      (5)      (6)
## -----
## treat                               -8.454                               -4.248
##                               t = -0.854                               t = -0.359
##
## after          -13.875*   -13.875*   7.159***                               -2.105
##                               t = -1.719 t = -1.720 t = 6.559                               t = -0.412
##
## treatTRUE:after          -21.034*** -21.364*** -11.771   -11.771
##                               t = -2.721 t = -2.764 t = -1.267 t = -1.268
##
## Constant          99.254***          107.708***          103.501***
##                               t = 9.546          t = 115.757          t = 16.295
##
## -----
## Observations          830          830          250,359   250,359   3,320   3,320
```

```
## =====
## Note:                                     *p<0.1; **p<0.05; ***p<0.01
```